

Example: MayerVietoris Patching of Broken Cohomology Classes

Setting

Let $X = \mathbb{P}^2$, the complex projective plane. Cover X with two affine open sets:

- $U = \{[x : y : z] \in \mathbb{P}^2 \mid x \neq 0\}$
- $V = \{[x : y : z] \in \mathbb{P}^2 \mid y \neq 0\}$

Then $U \cup V = X$ and $U \cap V$ is the subset where both $x \neq 0$ and $y \neq 0$.

Step 1: Local Classes

Let us consider cohomology classes in degree 2 (i.e., H^2). In this case:

$$H^2(X, \mathbb{Q}) \cong \mathbb{Q} \quad (\text{generated by the hyperplane class})$$

Let $\alpha_U \in H^2(U, \mathbb{Q})$, $\alpha_V \in H^2(V, \mathbb{Q})$, such that both represent local approximations to the global hyperplane class, but they do not agree exactly on the overlap $U \cap V$.

Step 2: MayerVietoris Sequence

We apply the long exact sequence:

$$\cdots \rightarrow H^1(U \cap V) \xrightarrow{\delta} H^2(X) \rightarrow H^2(U) \oplus H^2(V) \rightarrow H^2(U \cap V) \rightarrow \cdots$$

Let:

$$(\alpha_U, \alpha_V) \mapsto \alpha_{UV} = \alpha_U|_{U \cap V} - \alpha_V|_{U \cap V}$$

Suppose $\alpha_{UV} \neq 0$. Then α_U and α_V are mismatched — this is a **broken matrix** situation.

Step 3: Repairing the Class

The mismatch $\alpha_{UV} \in H^2(U \cap V)$ is sent backward through δ into $H^3(X)$, which is trivial for $\mathbb{P}^2$. Thus, the image is zero, and the pair (α_U, α_V) does glue to a global class in $H^2(X)$.

Even though locally broken, *the cohomology class globally resolves* because the obstruction vanishes.

Step 4: Lifting Algebraic Cycles

If α_U, α_V arise from algebraic divisors locally, then by MayerVietoris:

- Their mismatch is a computable obstruction.
- If this obstruction maps to zero, a global algebraic cycle exists.

Thus, you achieve a recursive lifting: local algebraicity patched through MayerVietoris yields global algebraicity.